



DC – AI – Routing – Switching – Security – Wireless – Automation

ACX7100 Routers Deepdive

An HPE Juniper Networking Techpost

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HPE



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1 ACX7100 Deepdive

Pankaj Kumar - 08/16/2022

In this post, you'll learn everything about the first two Juniper Cloud-Metro devices: ACX7100-32C and ACX7100-48L.

Article co-written by Nicolas Fevrier and Pankaj Kumar.

1.1 High-Level Introduction

ACX7100 is the first Juniper product powered by Broadcom Jericho2 [BCM88690] and running Junos EVO modular software architecture.

The products exists in two flavors:

- ACX7100-48L: 48 ports 10/25/50GE and 6 ports 100/200/400GE
- ACX7100-32C: 32 ports 40/100GE and 4 ports 100/200/400GE



Figure 1: ACX7100-48 Front View



Figure 2: ACX7100-32C Front View

Both devices are 1RU Ethernet-only routers with 4.8 Tbps forwarding capacity and Class-C Timing. MACsec is supported on all the ports of the ACX7100-32C only.

They are primarily targeted at the aggregation market but can be used in multiple roles once features and scales meet the requirements.

Maximum number of ports supported on ACX7100 Routers:

Port Speed	ACX7100-32CTotal Supported Ports	ACX7100-32CDetails	ACX7100-48LTotal Supported Ports	ACX7100-48LDetails
400GE	4	4x 1	6	6x 1
100GE	48	32x 1 + 4x (4x1)	24	6x (4x1)
50GE	96	32x (2x1) + 4x (8x1)	95	47 + 6x (8x1)

Port Speed	ACX7100-32CTotal Supported Ports	ACX7100-32CDetails	ACX7100-48LTotal Supported Ports	ACX7100-48LDetails
40GE	36	32x 1 + 4x 1	6	6x 1
25GE	80	16x (4x1) + 4x (4x1)	72	48 + 6x (4x1)
10GE	80	16x (4x1) + 4x (4x1)	72	48 + 6x (4x1)
1GE	0*	0*	0*	0*

At the moment of this publication, solutions are under investigation to offer 1GE connectivity.

1.1.1 Naming convention

In ACX7100 systems, aside from optics, the only field replaceable parts are the power supply modules, the fan trays and the SATA SSD.

We have a total of 6 SKUs available based on three parameters:

- Port density and port type: 32C for 32x 100GE and 48L for 48x (10/25/50GE)
- Power supply: AC or DC PSM
- Airflow: AFO (Air Flow Out / front-to-back) or AFI (Air Flow In / back-to-front)

Based on these parameters, the naming logic for the product itself and the SKU is the following.



Figure 3: 32C Model



Figure 4: 48L Model

SKUs:

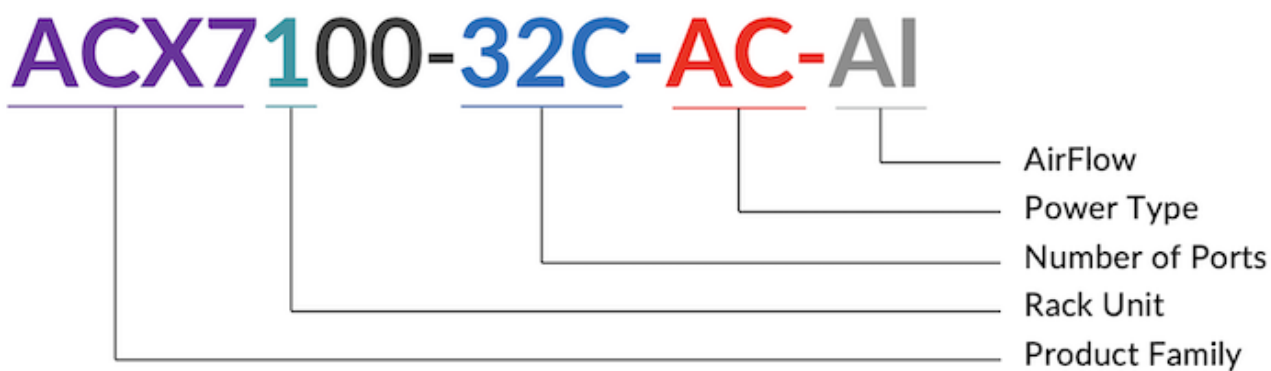


Figure 5: SKU information

The ACX7100-48L exists in both AFO (front to back) and AFI (back to front) versions:

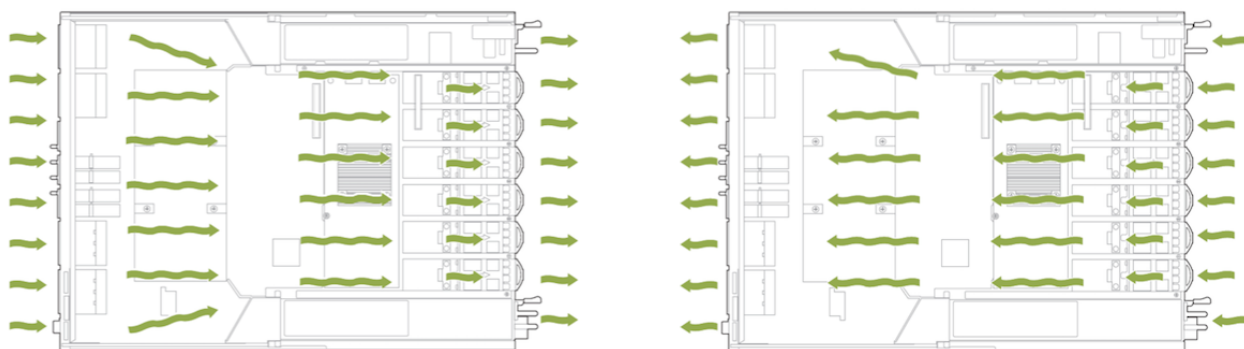
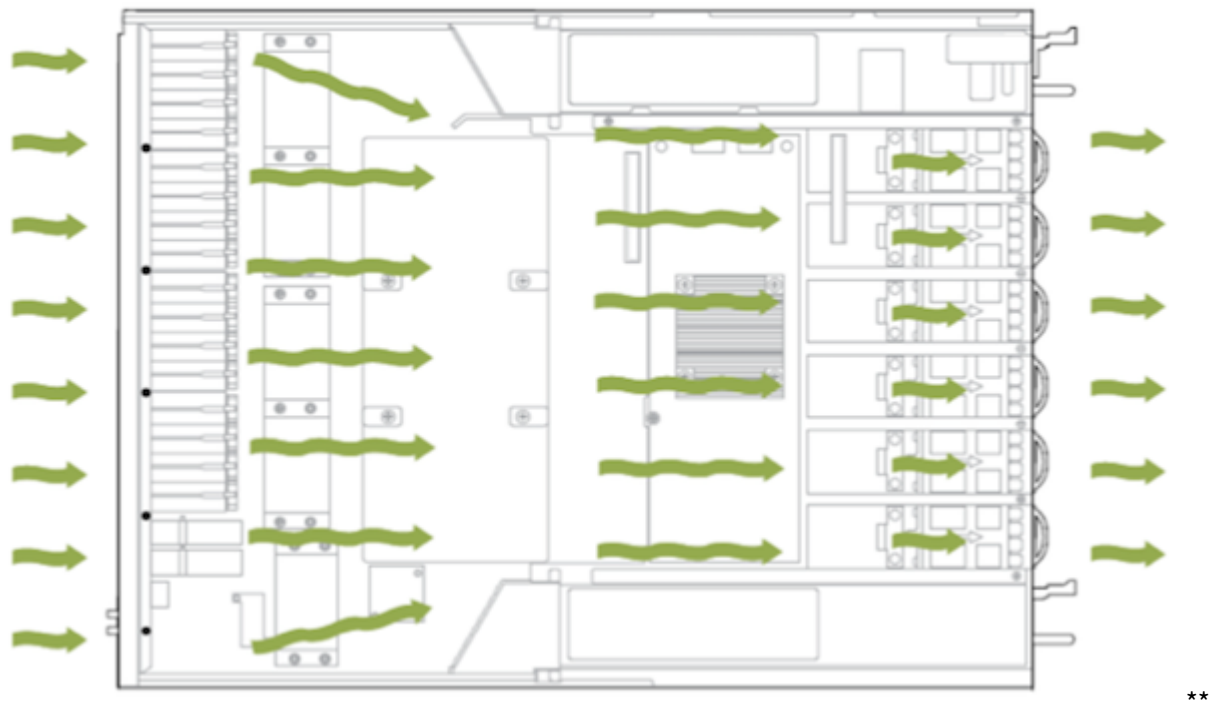


Figure 6: Version information

The ACX7100-32C is only available in AFO version:



It gives us a total of 6 SKUs:

SKU	Description
ACX7100-48L-AC-AO	48x 50GE + 6x 400GE / AC PSM / Front-to-Back cooling
ACX7100-48L-AC-AI	48x 50GE + 6x 400GE / AC PSM / Back-to-Front cooling
ACX7100-48L-DC-AO	48x 50GE + 6x 400GE / DC PSM / Front-to-Back cooling
ACX7100-48L-DC-AI	48x 50GE + 6x 400GE / DC PSM / Back-to-Front cooling
ACX7100-32C-AC-AO	32x 100GE + 4x 400GE / AC PSM / Front-to-Back cooling
ACX7100-32C-DC-AO	32x 100GE + 4x 400GE / DC PSM / Front-to-Back cooling

Note: we can **not** mix Fan Tray types (AFI+AFO) or PSM units (AC+DC).

We invite you to take a look at the TechLibrary hardware guide for more details: - <https://www.juniper.net/documentation/us/en/hardware/acx7100-48L/topics/topic-map/acx7100-48L-system-overview.html> - <https://www.juniper.net/documentation/us/en/hardware/acx7100-32C/topics/topic-map/acx7100-32C-system-overview.html>

1.1.2 ACX7100-48L

```

1 regress@rtme-acx-48L> show chassis hardware
2
3 Hardware inventory:
4 Item Version Part number Serial number Description
5 Chassis YW02210xxxxx JNP7100-48L [ACX7100-48L]
6 PSM 0 REV 04 740-085431 1ED7A1xxxxx PSU 1600W AC, Front to Back Airflow
7 PSM 1 REV 04 740-085431 1ED7A1xxxxx PSU 1600W AC, Front to Back Airflow

```

```

8 Routing Engine 0 REV 08 611-112446 YY02210xxxxx RE-JNP-7100
9 CB 0 REV 10 650-113149 YW02210xxxxx Control Board
10 FPC 0 BUILTIN BUILTIN ACX7100-48L
11 PIC 0 BUILTIN BUILTIN MRATE- 48xSFP56 + 6xQSFP56-DD
12 Xcvr 0 REV 01 740-021308 97D625Axxxxx SFP+-10G-SR
13 Xcvr 1 REV 01 740-021308 AJQxxxxx SFP+-10G-SR
14 Xcvr 3 0 NON-JNPR ONT154xxxxx SFP+-10G-LR
15 Xcvr 4 REV 01 740-031981 AP4xxxxx SFP+-10G-LR
16 Xcvr 6 REV 01 740-031981 45T0124xxxxx SFP+-10G-LR
17 Xcvr 7 REV 01 740-031981 38T0124xxxxx SFP+-10G-LR
18 Xcvr 42 REV 01 740-031981 45T0124xxxxx SFP+-10G-LR
19 Xcvr 43 0 NON-JNPR ONT154xxxxx SFP+-10G-LR
20 Xcvr 45 REV 01 740-021308 ALDxxxxx SFP+-10G-SR
21 Xcvr 48 REV 01 740-058734 1F1CQ1Axxxxxx QSFP-100GBASE-SR4
22 Xcvr 50 XXXX NON-JNPR L2128xxxxx QSFP56-DD-400G-ZR
23 Xcvr 51 XXXX NON-JNPR L2131xxxxx QSFP56-DD-400G-ZR
24 Fan Tray 0 JNP7100 Fan Tray, Front to Back Airflow - AFO
25 Fan Tray 1 JNP7100 Fan Tray, Front to Back Airflow - AFO
26 Fan Tray 2 JNP7100 Fan Tray, Front to Back Airflow - AFO
27 Fan Tray 3 JNP7100 Fan Tray, Front to Back Airflow - AFO
28 Fan Tray 4 JNP7100 Fan Tray, Front to Back Airflow - AFO
29 Fan Tray 5 JNP7100 Fan Tray, Front to Back Airflow - AFO
30
31 regress@rtme-acx-48L>

```

1.1.3 ACX7100-32C

```

1 regress@rtme-acx7100-32C> show chassis hardware
2
3 Hardware inventory:
4 Item Version Part number Serial number Description
5 Chassis YX02211xxxxx JNP7100-32C [ACX7100-32C]
6 PSM 0 REV 04 740-085431 1ED7A1xxxxxx PSU 1600W AC, Front to Back Airflow
7 PSM 1 REV 04 740-085431 1ED7A1xxxxxx PSU 1600W AC, Front to Back Airflow
8 Routing Engine 0 REV 09 611-112446 YY02211xxxxx RE-JNP-7100
9 CB 0 REV 13 650-113148 YX02211xxxxx Control Board
10 FPC 0 BUILTIN BUILTIN ACX7100-32C
11 PIC 0 BUILTIN BUILTIN MRATE- 32xQSFP-DD + 4xQSFP56-DD
12 Xcvr 0 REV 01 740-058734 1A3CQ1A5xxxxxx QSFP-100GBASE-SR4
13 Xcvr 1 REV 01 740-058734 1F1CQ1A6xxxxxx QSFP-100GBASE-SR4
14 Xcvr 2 REV 01 740-058734 1F1CQ1A6xxxxxx QSFP-100GBASE-SR4
15 Xcvr 3 REV 01 740-058734 1F1CQ1A6xxxxxx QSFP-100GBASE-SR4
16 Xcvr 4 REV 01 740-058734 1F1CQ1A6xxxxxx QSFP-100GBASE-SR4
17 Xcvr 5 REV 01 740-058734 1F1CQ1A6xxxxxx QSFP-100GBASE-SR4
18 Fan Tray 0 JNP7100 Fan Tray, Front to Back Airflow - AFO
19 Fan Tray 1 JNP7100 Fan Tray, Front to Back Airflow - AFO
20 Fan Tray 2 JNP7100 Fan Tray, Front to Back Airflow - AFO
21 Fan Tray 3 JNP7100 Fan Tray, Front to Back Airflow - AFO
22 Fan Tray 4 JNP7100 Fan Tray, Front to Back Airflow - AFO
23 Fan Tray 5 JNP7100 Fan Tray, Front to Back Airflow - AFO
24
25 regress@rtme-acx7100-32C>

```

1.1.4 Software

The minimum Junos release required to operate ACX7100 is 21.1R1.

The image is named: junos-evo-install-acx-f-x86-64-EVO

```

1 regress@rtme-acx7100-32> show version
2
3 Hostname: rtme-acx7100-32C
4 Model: ACX7100-32C
5 Junos: 22.2R1.12-EVO
6 Yocto: 3.0.2
7 Linux Kernel: 5.2.60-yocto-standard-g34a7a87
8 JUNOS-EVO OS 64-bit [junos-evo-install-acx-f-x86-64-22.2R1.12-EVO]
9
10 regress@rtme-acx7100-32C>

```

```

1 regress@rtme-acx-48L> show version

```



```

2
3 Hostname: rtme-acx-48L
4 Model: ACX7100-48L
5 Junos: 22.2R1.12-EVO
6 Yocto: 3.0.2
7 Linux Kernel: 5.2.60-yocto-standard-g34a7a87
8 JUNOS-EVO OS 64-bit junos-evo-install-acx-f-x86-64-22.2R1.12-EVO]
9
10 regress@rtme-acx-48L>

```

Both ACX7100-32C and ACX7100-48L are using the same “acx-f” images (f for fixed). “acx-m” is used for modular systems.

1.2 ACX7100 Ethernet Ports

To verify the latest support of optics, always use the Hardware Compatibility Tool: <https://apps.juniper.net/hct/product/?prd=ACX7100-48L> and <https://apps.juniper.net/hct/product/?prd=ACX7100-32C>

In ACX7k Series, all physical interfaces are named “et-*” regardless of the optics/speed. It simplifies significantly the port migration tasks. In this example below, you can verify that port 10GE, 100GE or 400GE are all “et-x/y/z”:

```

1 regress@rtme-acx-48L> show interfaces et-0/0/0
2
3 Physical interface: et-0/0/0, Enabled, Physical link is Up
4   Interface index: 1008, SNMP ifIndex: 505
5   Link-level type: Ethernet, MTU: 1514, LAN-PHY mode, Speed: 10Gbps, BPDU Error: None, Loop
      Detect PDU Error: None, Ethernet-Switching Error: None, MAC-REWRITE Error: None, Loopback:
      Disabled,
6   Source filtering: Disabled, Flow control: Enabled, Auto-negotiation: Disabled, Media type:
      Fiber
7   Device flags : Present Running
8   Interface flags: SNMP-Traps
9   CoS queues : 8 supported, 8 maximum usable queues
10  Current address: ac:78:d1:23:a3:a8, Hardware address: ac:78:d1:23:a3:a8
11  Last flapped : 2022-08-12 20:27:57 PDT (4d 14:14 ago)
12  Input rate : 0 bps (0 pps)
13  Output rate : 0 bps (0 pps)
14  Active alarms : None
15  Active defects : None
16  PCS statistics Seconds
17    Bit errors 0
18    Errored blocks 0
19  Ethernet FEC Mode : NONE
20  Ethernet FEC statistics Errors
21    FEC Corrected Errors 0
22    FEC Uncorrected Errors 0
23    FEC Corrected Errors Rate 0
24    FEC Uncorrected Errors Rate 0
25  Interface transmit statistics: Disabled
26  Link Degradation :
27    Link Monitoring : Disable
28
29  Logical interface et-0/0/0.16386 (Index 1008) (SNMP ifIndex 613)
30    Flags: Up SNMP-Traps Encapsulation: ENET2
31    Input packets : 0
32    Output packets: 0
33    Protocol multiservice, MTU: Unlimited
34    Flags: None
35
36 regress@rtme-acx-48L> show interfaces et-0/0/48
37
38 Physical interface: et-0/0/48, Enabled, Physical link is Down
39   Interface index: 1056, SNMP ifIndex: 601
40   Link-level type: Ethernet, MTU: 1514, LAN-PHY mode, Speed: 100Gbps, BPDU Error: None, Loop
      Detect PDU Error: None, Ethernet-Switching Error: None, MAC-REWRITE Error: None, Loopback:
      Disabled,
41   Source filtering: Disabled, Flow control: Enabled, Auto-negotiation: Disabled, Media type:
      Fiber
42   Device flags : Present Running Down
43   Interface flags: Hardware-Down SNMP-Traps
44   CoS queues : 8 supported, 8 maximum usable queues

```

```

45 Current address: ac:78:d1:23:a5:28, Hardware address: ac:78:d1:23:a5:28
46 Last flapped : Never
47 Input rate : 0 bps (0 pps)
48 Output rate : 0 bps (0 pps)
49 Active alarms : LINK
50 Active defects : LINK, LOCAL-FAULT
51 PCS statistics Seconds
52   Bit errors 0
53   Errored blocks 0
54 Ethernet FEC Mode : FEC91
55 Ethernet FEC statistics Errors
56   FEC Corrected Errors 0
57   FEC Uncorrected Errors 0
58   FEC Corrected Errors Rate 0
59   FEC Uncorrected Errors Rate 0
60 Interface transmit statistics: Disabled
61 Link Degradate :
62   Link Monitoring : Disable
63
64 Logical interface et-0/0/48.0 (Index 1015) (SNMP ifIndex 512)
65   Flags: Up SNMP-Traps Encapsulation: ENET2
66   Input packets : 0
67   Output packets: 0
68   Protocol inet
69   Max nh cache: 100000, New hold nh limit: 100000, Curr nh cnt: 0, Curr new hold cnt: 0, NH
   drop cnt: 0
70   MTU: 1500
71   Flags: Sendbcst-pkt-to-re
72   Addresses, Flags: Is-Preferred Is-Primary
73     Destination: 23.23.23.0/30.0, Local: 23.23.23.1, Broadcast: 23.23.23.3
74   Protocol multiservice, MTU: Unlimited
75   Flags: None
76
77 regress@rtme-acx-48L> show interfaces et-0/0/49
78
79 Physical interface: et-0/0/49, Enabled, Physical link is Down
80 Interface index: 1057, SNMP ifIndex: 603
81 Link-level type: Ethernet, MTU: 1514, LAN-PHY mode, Speed: 400Gbps, BPDU Error: None, Loop
   Detect PDU Error: None, Ethernet-Switching Error: None, MAC-REWRITE Error: None, Loopback:
   Disabled,
82 Source filtering: Disabled, Flow control: Enabled, Auto-negotiation: Disabled, Media type:
   Fiber
83 Device flags : Present Running Down
84 Interface flags: Hardware-Down SNMP-Traps
85 CoS queues : 8 supported, 8 maximum usable queues
86 Current address: ac:78:d1:23:a5:30, Hardware address: ac:78:d1:23:a5:30
87 Last flapped : Never
88 Input rate : 0 bps (0 pps)
89 Output rate : 0 bps (0 pps)
90 Active alarms : LINK
91 Active defects : LINK, LOCAL-FAULT
92 PCS statistics Seconds
93   Bit errors 0
94   Errored blocks 0
95 Ethernet FEC Mode : FEC119
96 Ethernet FEC statistics Errors
97   FEC Corrected Errors 0
98   FEC Uncorrected Errors 0
99   FEC Corrected Errors Rate 0
100  FEC Uncorrected Errors Rate 0
101 Interface transmit statistics: Disabled
102 Link Degradate :
103   Link Monitoring : Disable
104
105 Logical interface et-0/0/49.16386 (Index 1055) (SNMP ifIndex 657)
106   Flags: Up SNMP-Traps Encapsulation: ENET2
107   Input packets : 0
108   Output packets: 0
109   Protocol multiservice, MTU: Unlimited
110   Flags: None
111
112 regress@rtme-acx-48L>

```

1.2.1 ACX7100-48L Front Panel view and Port Numbering

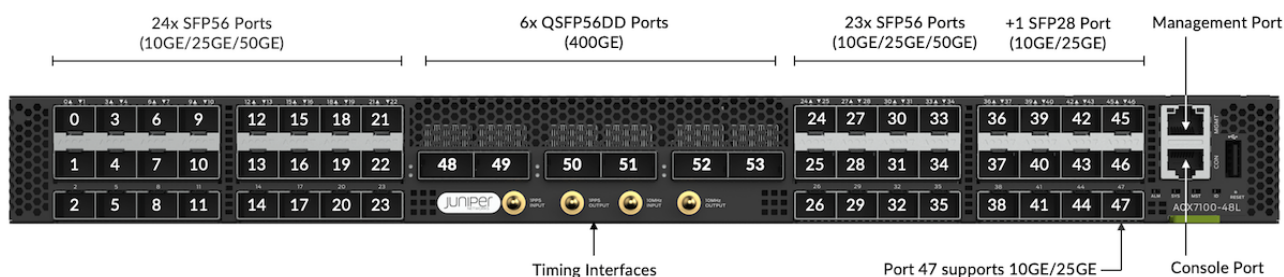


Figure 7: Port Numbering

Port Number(s)	Optic Type	Supported Ports	Default Port Type
0 - 46	SFP+, SFP28, SFP56	10GE, 25GE, 50GE	1x10GE
47	SFP+ or SFP28	10GE, 25GE	1x10GE
48 - 53	QSFP56-DD	100GE, 200GE, 400GE	1x400GE

We have various breakout options available like 4x10GE, 4x25GE, 2x50GE, 8x50GE along with native 10GE, 25GE, 50GE, 100GE and 400GE support.

Check the port capabilities with the CLI command `show chassis pic fpc-slot 0 pic-slot 0`.

```

1 regress@rtme-acx-L> show chassis pic fpc-slot 0 pic-slot 0
2
3 FPC slot 0, PIC slot 0 information:
4   Type MRATE- 48xSFP56 + 6xQSFP56-DD
5
6 <SNIP>
7
8 Port speed information:
9
10  Port PFE Capable Port Speeds
11  0 0 1x10G 1x25G 1x50G
12  1 0 1x10G 1x25G 1x50G
13  2 0 1x10G 1x25G 1x50G
14  3 0 1x10G 1x25G 1x50G
15  4 0 1x10G 1x25G 1x50G
16  5 0 1x10G 1x25G 1x50G
17  6 0 1x10G 1x25G 1x50G
18  7 0 1x10G 1x25G 1x50G
19  8 0 1x10G 1x25G 1x50G
20  9 0 1x10G 1x25G 1x50G
21 10 0 1x10G 1x25G 1x50G
22 11 0 1x10G 1x25G 1x50G
23 12 0 1x10G 1x25G 1x50G
24 13 0 1x10G 1x25G 1x50G
25 14 0 1x10G 1x25G 1x50G
26 15 0 1x10G 1x25G 1x50G
27 16 0 1x10G 1x25G 1x50G
28 17 0 1x10G 1x25G 1x50G
29 18 0 1x10G 1x25G 1x50G
30 19 0 1x10G 1x25G 1x50G
31 20 0 1x10G 1x25G 1x50G
32 21 0 1x10G 1x25G 1x50G
33 22 0 1x10G 1x25G 1x50G
34 23 0 1x10G 1x25G 1x50G
35 24 0 1x10G 1x25G 1x50G
36 25 0 1x10G 1x25G 1x50G
37 26 0 1x10G 1x25G 1x50G
38 27 0 1x10G 1x25G 1x50G
39 28 0 1x10G 1x25G 1x50G
40 29 0 1x10G 1x25G 1x50G

```

```

41 30 0 1x10G 1x25G 1x50G
42 31 0 1x10G 1x25G 1x50G
43 32 0 1x10G 1x25G 1x50G
44 33 0 1x10G 1x25G 1x50G
45 34 0 1x10G 1x25G 1x50G
46 35 0 1x10G 1x25G 1x50G
47 36 0 1x10G 1x25G 1x50G
48 37 0 1x10G 1x25G 1x50G
49 38 0 1x10G 1x25G 1x50G
50 39 0 1x10G 1x25G 1x50G
51 40 0 1x10G 1x25G 1x50G
52 41 0 1x10G 1x25G 1x50G
53 42 0 1x10G 1x25G 1x50G
54 43 0 1x10G 1x25G 1x50G
55 44 0 1x10G 1x25G 1x50G
56 45 0 1x10G 1x25G 1x50G
57 46 0 1x10G 1x25G 1x50G
58 47 0 1x10G 1x25G
59 48 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
60 49 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
61 50 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
62 51 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
63 52 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
64 53 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
65 54 NA 1x10G
66
67 regress@rtme-acx-48L>

```

Port 54 is internal-only and used to connect to the PTP (Precision Time Protocol) FPGA via a mux. You'll also note that ACX7100-48L doesn't support 1GE ports at FRS (launch date). A solution to provide 1GE capability is under consideration, but it's not a mode natively supported by Jericho2.

1.2.2 ACX7100-48L Port Groups

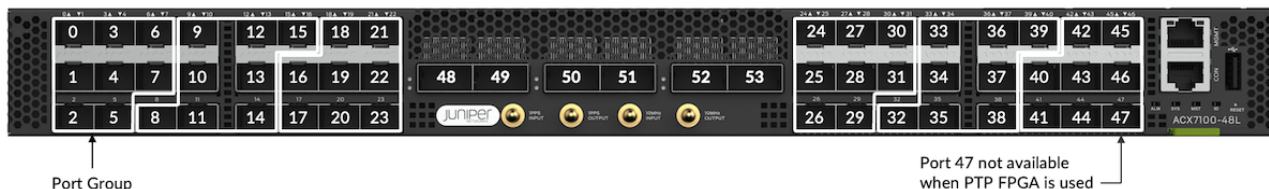


Figure 8: ACX7100-48L Port Groups and Their Interfaces*

To understand the configuration capabilities, we need to introduce the concept of “Port Group” (PG). As the name implies, a PG gathers several adjacent ports.

Internally, the ports of a given PG are mapped to the same Port Macro (PM). A PM is block of SerDes (Serializer/Deserializer) connecting the PFE to other elements inside the router, including Network Interfaces and Fabric Ports. For more details on this concept, we invite you to read the “Port Macros” section of this article: <https://juniper.github.io/techposts/building-the-acx7000-series-the-pfe/article>

We can abstract it to this diagram:

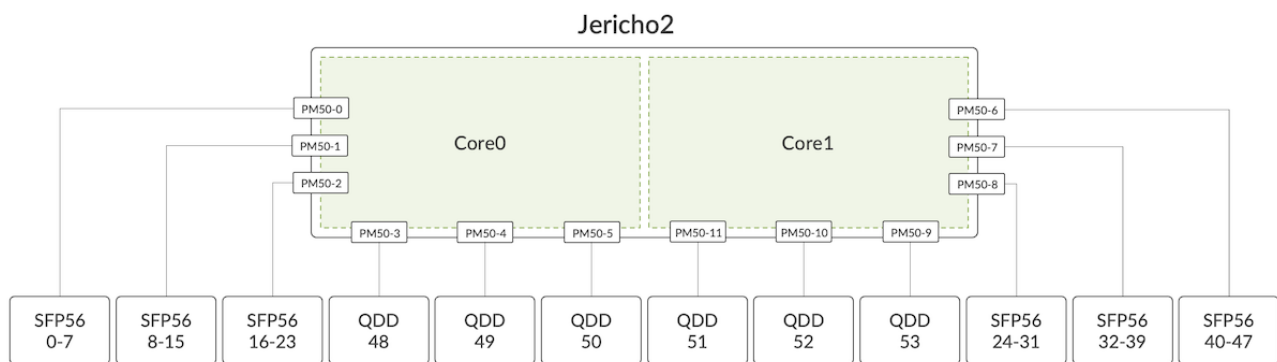


Figure 9: Mapping of ports to Port Macro and J2 Cores

For the SFP ports, from 0 to 47, 8 ports constitute a group.

PG 0	Ports 0 - 7
PG 1	Ports 8 - 15
PG 2	Ports 16 - 23
PG 3	Ports 24 - 31
PG 4	Ports 32 - 39
PG 5	Ports 40 - 47

Depending on their electrical specifications, optic modules require appropriate clock sources. A given Port Group can only be configured with two clock speeds. It impacts the combination of port types we support within this PG. For example, it will accept 10 Gbps + 25 Gbps, or 10 Gbps + 50 Gbps, or 25 Gbps + 50 Gbps, but it won't be possible to add a third clock speed in the mix.

The QSFP ports from 48 to 53 are individually considered as a Port Group, there are no specific limitations for them.

1.2.3 ACX7100-48L: Port Configuration and Port Checker Tool

Always use the port checker tool to verify the support and understand the potential restrictions: <https://apps.juniper.net/home/port-checker/index.html>

These limitations in port usage and port combination are the following:

Once PTP is enabled, Port 47 is not usable

This port 47 can be used with 10GE or 25GE, but not 50GE optics, that's why you'll find in the public documentation that router forwarding capability is 4.775 Tbps (and not 4.8T). Once the PTP FPGA is enabled, the port is disabled. Indeed, we use an internal mux to connect to the PTP chipset.

To activate the PTP FPGA, use the following CLI configuration:

```
1 set chassis fpc 0 pic 0 ptp-mode
```

Port Checker

MX204
MX10003
MX304
ACX7100-32C
ACX7100-48L
ACX7509

ACX7100-48L

PTP FPGA

PTP enabled

Port 0 Empty	Port 3 Empty	Port 6 Empty	Port 9 Empty	Port 12 Empty	Port 15 Empty	Port 18 Empty	Port 21 Empty	Port 24 Empty	Port 27 Empty	Port 30 Empty	Port 33 Empty	Port 36 Empty	Port 39 Empty	Port 42 Empty	Port 45 Empty
Port 1 Empty	Port 4 Empty	Port 7 Empty	Port 10 Empty	Port 13 Empty	Port 16 Empty	Port 19 Empty	Port 22 Empty	Port 25 Empty	Port 28 Empty	Port 31 Empty	Port 34 Empty	Port 37 Empty	Port 40 Empty	Port 43 Empty	Port 46 Empty
Port 2 Empty	Port 5 Empty	Port 8 Empty	Port 11 Empty	Port 14 Empty	Port 17 Empty	Port 20 Empty	Port 23 Empty	Port 26 Empty	Port 29 Empty	Port 32 Empty	Port 35 Empty	Port 38 Empty	Port 41 Empty	Port 44 Empty	Port 47 Empty
Port 48 Empty	Port 49 Empty	Port 50 Empty	Port 51 Empty	Port 52 Empty	Port 53 Empty										

Port 47 not available

```

1 [edit]
2 root@rtme-acx7100# run show interfaces terse et-0/0/6*
3 Interface Admin Link Proto Local Remote
4 et-0/0/6 up down
5 et-0/0/6.16386 up down multiservice
6
7 [edit]
8 root@rtme-acx7100# run show interfaces terse et-0/0/7*
9 Interface Admin Link Proto Local Remote
10 et-0/0/7 up down
11 et-0/0/7.16386 up down multiservice
12
13 [edit]
14 root@rtme-acx7100# show interfaces et-0/0/6
15
16 [edit]
17 root@rtme-acx7100# show interfaces et-0/0/7
18
19 [edit]
20 root@rtme-acx7100# set interfaces et-0/0/6 number-of-sub-ports 4 speed 10g
21
22 [edit]
23 root@rtme-acx7100# show interfaces et-0/0/6
24 number-of-sub-ports 4;
25 speed 10g;
26
27 [edit]
28 root@rtme-acx7100# commit
29 commit complete
30
31 [edit]
32 root@rtme-acx7100# run show interfaces terse et-0/0/6*
33 Interface Admin Link Proto Local Remote
34 et-0/0/6 up down
35 et-0/0/6.16386 up down multiservice
36
37 [edit]
38 root@rtme-acx7100# run show interfaces terse et-0/0/7*
39 Interface Admin Link Proto Local Remote
40 et-0/0/7 up down
41 et-0/0/7.16386 up down multiservice
42
43 [edit]
44 root@rtme-acx7100# set interfaces et-0/0/7 unused
45
46 [edit]
47 root@rtme-acx7100# commit
48 commit complete
49
50 [edit]
51 root@rtme-acx7100# run show interfaces terse et-0/0/6*
52 Interface Admin Link Proto Local Remote
53 et-0/0/6:0 up down
54 et-0/0/6:1 up down
55 et-0/0/6:2 up down
56 et-0/0/6:3 up down
57

```

```

58 [edit]
59 root@rtme-acx7100# run show interfaces terse et-0/0/7*
60 error: device et-0/0/7* not found
61
62 [edit]
63 root@rtme-acx7100#

```

Only two SerDes speeds can be configured in a given Port Group

Keep in mind that port speed and SerDes rates are different, the second category is Broadcom internal rates: 10.3125Gbps, 20.725Gbps, 25.7825Gbps, 53.125Gbps...

SFP56Ports 0 to 47	J2 SerDes(Gbps)
50GE (not supported in port 47)	53.125
25GE	25.7825
10GE	10.3125
QSFP56-DDPorts 32 to 35	J2 SerDes(Gbps)
400GE	8x 53.125
4x 100GE BreakOut	8x 53.125
2x 100GE BreakOut	8x 25.7825
100GE (SR4/LR4/ER4/...)	4x 25.7825

Port Checker

MX204 MX10003 MX304 ACX7100-32C **ACX7100-48L** ACX7509

ACX7100-48L

☐ PTP FPGA **PG-0**

In each Port Group (PG-0 made of ports 0-7) we can only mix two port speeds out of 10G, 25G and 50G

Port 0 50GE	Port 3 Empty	Port 6 Empty	Port 9 Empty	Port 12 Empty	Port 15 Empty	Port 18 Empty	Port 21 Empty	Port 24 Empty	Port 27 Empty	Port 30 Empty	Port 33 Empty	Port 36 Empty	Port 39 Empty	Port 42 Empty	Port 45 Empty
Port 1 Empty	Port 4 25GE	Port 7 Empty	Port 10 Empty	Port 13 Empty	Port 16 Empty	Port 19 Empty	Port 22 Empty	Port 25 Empty	Port 28 Empty	Port 31 Empty	Port 34 Empty	Port 37 Empty	Port 40 Empty	Port 43 Empty	Port 46 Empty
Port 2 Empty	Port 5 10GE	Port 8 Empty	Port 11 Empty	Port 14 Empty	Port 17 Empty	Port 20 Empty	Port 23 Empty	Port 26 Empty	Port 29 Empty	Port 32 Empty	Port 35 Empty	Port 38 Empty	Port 41 Empty	Port 44 Empty	Port 47 Empty
Port 48 Empty	Port 49 Empty	Port 50 Empty	Port 51 Empty	Port 52 Empty	Port 53 Empty										

ACX7100-48L front panel

Reason of invalid port selection

Configuration status: ❌

Reason: Invalid Configuration

- Configuration on ports 0, 1, 2, 3, 4, 5, 6 and 7 has exceeded the maximum SERDES rate on PFE (PG-0).

Figure 10: Port Checker

1.2.4 ACX7100-32C Front Panel View and Port Numbering

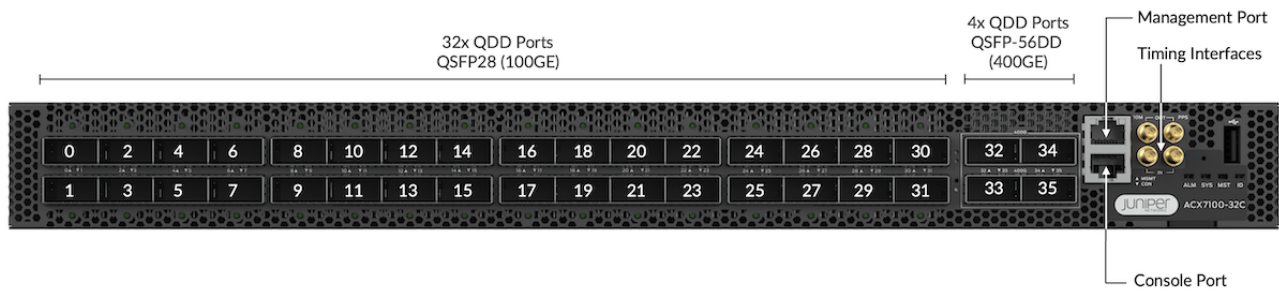


Figure 11: Port Numbering

Port Number(s)	Optic Type	Default Port Speed
0 - 31	QSFP-DD	1x100GE
32 - 35	QSFP56-DD	1x400GE

The first 32 ports are QSFP-DD cages mounted in a belly-to-belly fashion. With this design, they support high-power QSFP28 modules. A follow-up article will be published soon, demonstrating the use of ZR ports on all ports.

We also have various breakout options available like 4x10GE, 4x25GE, 2x50GE, 8x50GE along with native 10GE, 25GE, 50GE, 100GE and 400GE support.

Check the various port capabilities with the CLI command “show chassis pic fpc-slot 0 pic-slot 0”.

```

1 regress@rtme-acx7100-32C> show chassis pic fpc-slot 0 pic-slot 0
2
3 FPC slot 0, PIC slot 0 information:
4   Type MRATE- 32xQSFP-DD + 4xQSFP56-DD
5   State Online
6   PIC version 01.13
7   Uptime 20 days, 18 hours, 49 minutes, 7 seconds
8
9 <SNIP>
10
11 Port speed information:
12
13 Port PFE Capable Port Speeds
14 0 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
15 1 0 1x100G 1x40G 2x50G 1x25G 1x10G
16 2 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
17 3 0 1x100G 1x40G 2x50G 1x25G 1x10G
18 4 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
19 5 0 1x100G 1x40G 2x50G 1x25G 1x10G
20 6 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
21 7 0 1x100G 1x40G 2x50G 1x25G 1x10G
22 8 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
23 9 0 1x100G 1x40G 2x50G 1x25G 1x10G
24 10 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
25 11 0 1x100G 1x40G 2x50G 1x25G 1x10G
26 12 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
27 13 0 1x100G 1x40G 2x50G 1x25G 1x10G
28 14 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
29 15 0 1x100G 1x40G 2x50G 1x25G 1x10G
30 16 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
31 17 0 1x100G 1x40G 2x50G 1x25G 1x10G
32 18 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
33 19 0 1x100G 1x40G 2x50G 1x25G 1x10G
34 20 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
35 21 0 1x100G 1x40G 2x50G 1x25G 1x10G

```



```

36 22 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
37 23 0 1x100G 1x40G 2x50G 1x25G 1x10G
38 24 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
39 25 0 1x100G 1x40G 2x50G 1x25G 1x10G
40 26 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
41 27 0 1x100G 1x40G 2x50G 1x25G 1x10G
42 28 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
43 29 0 1x100G 1x40G 2x50G 1x25G 1x10G
44 30 0 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G
45 31 0 1x100G 1x40G 2x50G 1x25G 1x10G
46 32 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
47 33 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
48 34 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
49 35 0 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G
50 36 NA 1x10G
51
52 regress@rtme-acx7100-32C>

```

Note: port 36 is internal-only and used to connect to the PTP FPGA.

You can get similar info from the CLI-PFE by using the “show picd config” command (note that cli-pfe commands are not official supported and can change in their form or output without any notification).

```

1 [vrf:none] regress@rtme-acx7100-32C:~# cli-pfe
2
3 regress@rtme-acx7100-32C:pfe> show picd config
4
5
6 pic_info_table :
7     default
8
9     config config config computed
10 pic_or_port speed
11
12     pic_mode port_speed valid speed supported_speeds
13
14     hidden_speeds
15 -----
16
17 pic-0/0 - - - -
18 port-0/0/0 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
19 port-0/0/1 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
20 port-0/0/2 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
21 port-0/0/3 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
22 port-0/0/4 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
23 port-0/0/5 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
24 port-0/0/6 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
25 port-0/0/7 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
26 port-0/0/8 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
27 port-0/0/9 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
28 port-0/0/10 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
29 port-0/0/11 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
30 port-0/0/12 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
31 port-0/0/13 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
32 port-0/0/14 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
33 port-0/0/15 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
34 port-0/0/16 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
35 port-0/0/17 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
36 port-0/0/18 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
37 port-0/0/19 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
38 port-0/0/20 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
39 port-0/0/21 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
40 port-0/0/22 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
41 port-0/0/23 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
42 port-0/0/24 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
43 port-0/0/25 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
44 port-0/0/26 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
45 port-0/0/27 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
46 port-0/0/28 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
47 port-0/0/29 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
48 port-0/0/30 1x100G - - - 1x100G [ 1x100G 4x25G 1x40G 4x10G 2x50G 1x25G 1x10G ] [ ]
49 port-0/0/31 1x100G - - - 1x100G [ 1x100G 1x40G 2x50G 1x25G 1x10G ] [ ]
50 port-0/0/32 1x400G - - - 1x400G [ 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G ] [ ]
51 port-0/0/33 1x400G - - - 1x400G [ 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G ] [ ]
52 port-0/0/34 1x400G - - - 1x400G [ 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G ] [ ]

```

```

52 port-0/0/35 1x400G - - - 1x400G [ 1x400G 1x100G 1x40G 4x100G 2x100G 8x50G 2x50G 4x25G 4x10G ] [
53 ]
54 port-0/0/36 0x0 - - - 0x0 [ 1x10G ] [ ]
54 usr_config :
55
56 regress@rtme-acx7100-32C:pfe>

```

1.2.5 ACX7100-32C Port Groups

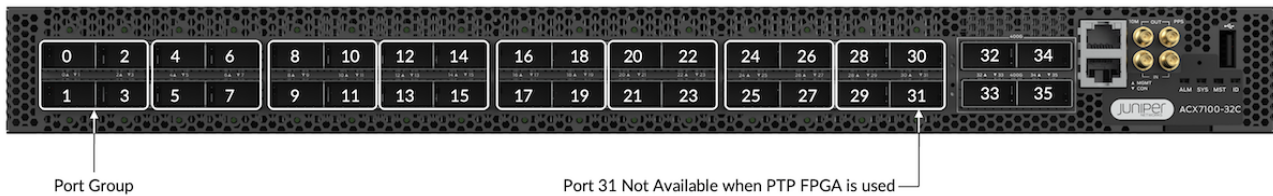
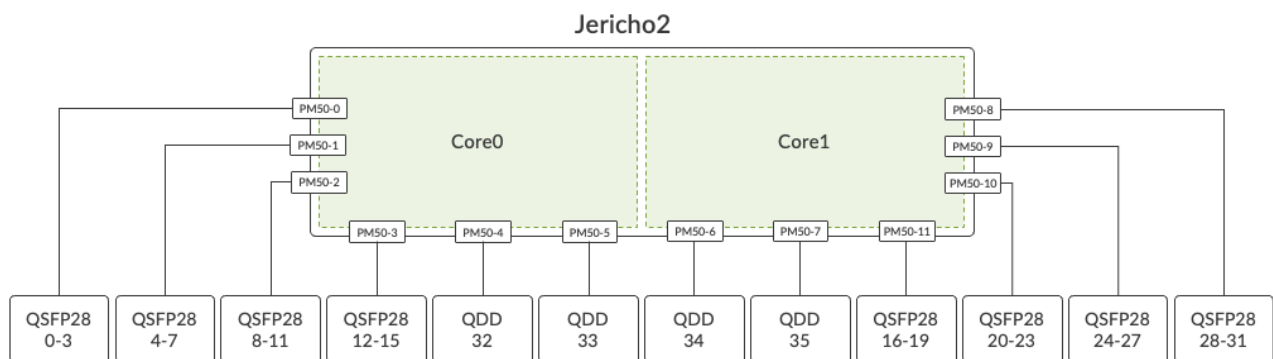


Figure 12: ACX7100-32C Port Groups and Their Interfaces

The same concept of Port Group described for ACX7100-48L exists in the ACX7100-32C. It influences the port configuration capabilities.



For the QSFP ports from 0 to 31, 4 subsequent ports constitute a group.

PG 0	Ports 0 - 3
PG 1	Ports 4 - 7
PG 2	Ports 8 - 11
PG 3	Ports 12 - 15
PG 4	Ports 16 - 19
PG 5	Ports 20 - 23
PG 6	Ports 24 - 27
PG 7	Ports 28 - 31

The QSFP56-DD ports from 32 to 35 are individually considered as a port group, so there are no specific limitations for them.

1.2.6 ACX7100-32C Port Configuration and Port Checker Tool

We invite you to use the port checker tool to verify the support and understand the potential restrictions: <https://apps.juniper.net/home/port-checker/index.html>

These limitations in port usage and port combination are the following:

Once PTP is enabled, Port 31 is not available (the same reason as port 47 on ACX7100-48L described in the previous section).

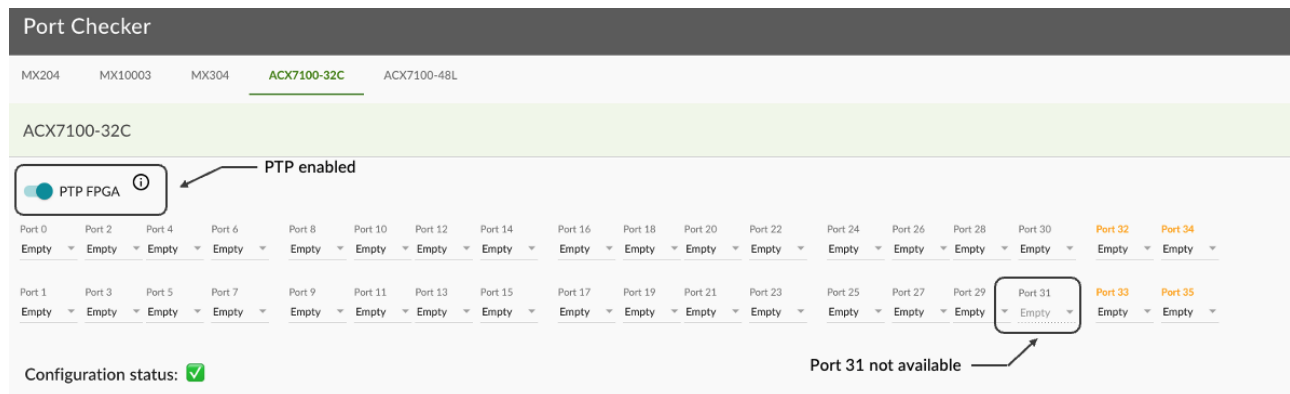


Figure 13: Port Checker 1

Only even ports can be channelized (configured with breakout). For odd ports, we don't allow the configuration of 4x10GE and 4x25GE as shown below.

Once an even port (N) is configured as channelized, the N+1 port must be configured as unused. This restriction is coming from the maximum number of internal MAC addresses we can use (logically, a break-out port consumes 4 MAC addresses).

Note: The ACX7100-48L supports 2x50GE on both odd and even ports, the limitation only applies to 4x10GE and 4x25GE.

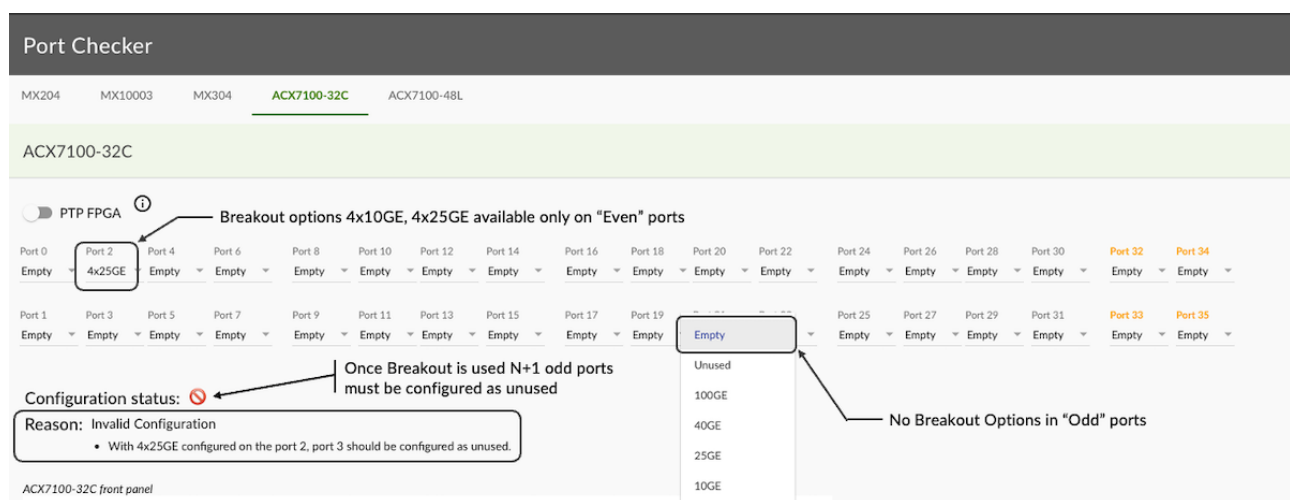


Figure 14: Port Checker 2

QSFP-DD Ports 0 to 31	J2 SerDes (Gbps)	Converted by GearBox into Line SerDes (Gbps)
100GE	2x 53.125	4x 25.7825
4x 25GE BreakOut	4x 25.7825	4x 25.7825
40GE	2x 20.725	4x 10.3125
4x 10GE BreakOut	4x 10.3125	4x 10.3125

QSFP56-DD Ports 32 to 35	J2 SerDes (Gbps)
400GE	8x 53.125
4x 100GE BreakOut	8x 53.125
2x 100GE BreakOut	8x 25.7825
100GE (SR4/LR4/ER4/...)	4x 25.7825

Port Checker

MX204 MX10003 MX304 **ACX7100-32C** ACX7100-48L User selecting three different SerDes rate in one port group

ACX7100-32C

PTP FPGA ⓘ

PG-2 ← In each Port Group (PG-2 made of ports 8-11) user selected three different SerDes rate

Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port	Port
Port 0	Port 2	Port 4	Port 6	Port 8	Port 10	Port 12	Port 14	Port 16	Port 18	Port 20	Port 22	Port 24	Port 26	Port 28	Port 30	Port 32	Port 34	Port 36	Port 38	
Empty	Empty	Empty	Empty	100GE	4x25GE	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	
Port 1	Port 3	Port 5	Port 7	Port 9	Port 11	Port 13	Port 15	Port 17	Port 19	Port 21	Port 23	Port 25	Port 27	Port 29	Port 31	Port 33	Port 35	Port 37	Port 39	
Empty	Empty	Empty	Empty	40GE	Unused	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	

Configuration status: ❌
Reason: Invalid Configuration
• Configuration on ports 8, 9, 10 and 11 has exceeded the maximum SERDES rate on PFE (PG-2).

Reason of invalid port selection

Figure 15: Port Checker 3

Note it's possible to use QSFP-to-SFP-Adaptors (QSA) in the 100GE slots. This use case could make sense if you need a small quantity of 10GE (particularly in long distances cases requiring ER reach, something not available with breakout options).

1.3 Internal Description: Block Diagrams

The ACX7100-32C is internally composed of multiple parts. We can summarize them as:

- A Broadcom Jericho2 PFE (BCM88690) with 8GB of HBM
- Multiple Gearboxes (GBs) to adapt 50 Gbps PAM4 into 25Gbps NRZ
- Multiple 400 Gbps PHY modules
- Intel x86 CPU (6 cores) with DDR4 memory and SSD storage
- Internal controllers for I2C (inter-integrated circuit), fan control, ...

- PTP FPGA
- TPM 2.0 module (Trusted Platform Module)

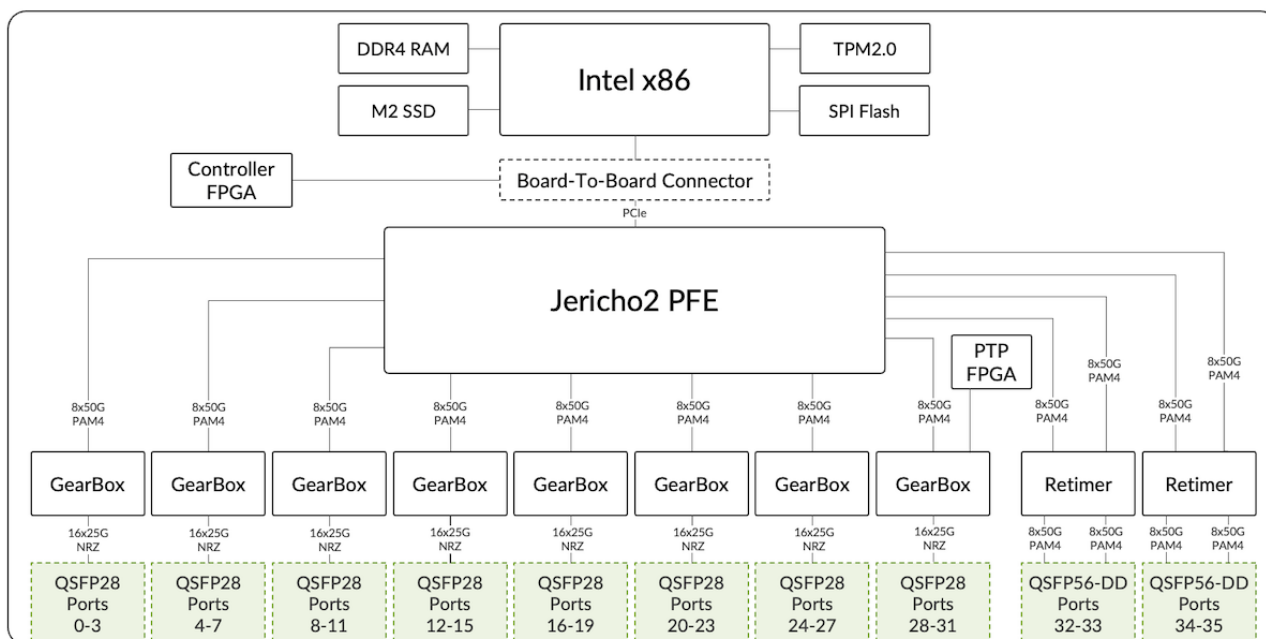


Figure 16: ACX7100-32C Block Diagram

In this diagram, we are presenting the SerDes at 50 Gbps (instead of 53.125 Gbps) and 25 Gbps (instead of 25.7875 Gbps) for sake of simplicity.

It is worth mentioning also that “NRZ” and “PAM4” mentioned below “GearBoxes” and “Retimers” in this diagram are dependent on the type of the optic modules inserted. Also, you could legitimately ask why we are using GB for the ports 32 to 35 since they could be directly connected to the Port Macros / PFE. This part is actually required for MACsec encryption, a technology not natively available in Jericho2 ASIC.

The different components can be monitored via CLI.

```

1 regress@rtme-acx7100-32C:pfe> show gearbox list fpc 0 pic 0
2
3 INDEX NAME MDIO INIT
4 -----
5 0 88X7121M_B0 0x0000 SUCCESS
6 1 88X7121M_B0 0x0008 SUCCESS
7 2 88X7121M_B0 0x0020 SUCCESS
8 3 88X7121M_B0 0x0028 SUCCESS
9 4 88X7121M_B0 0x0040 SUCCESS
10 5 88X7121M_B0 0x0048 SUCCESS
11 6 88X7121M_B0 0x0060 SUCCESS
12 7 88X7121M_B0 0x0068 SUCCESS
13 8 88X7121M_B0 0x0100 SUCCESS
14 9 88X7121M_B0 0x0108 SUCCESS
15 -----
16 API Version: 2.3.0
17 Serdes FW Rev: 0.10.11.89
18 Chip FW Rev: 0.2.16.1
19
20 regress@rtme-acx7100-32C:pfe> show gearbox port-status fpc 0 pic 0
21
22 PORT LINK(L:H) SPEED MODE(L:H) LANE_MAP(L:H) SD(L:H) LOCK(L:H) MDIO AN(L)
23 -----
24 0:0 --:UP 100G 62:69 0x000f:0x0003 0:1 0:1 0x0000(0) OFF
25 1:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0001(0) OFF
26 2:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x0002(0) OFF
27 3:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x0003(0) OFF
28 4:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0008(1) OFF
29

```

```

30 5:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0009(1) OFF
31 6:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x000a(1) OFF
32 7:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x000b(1) OFF
33 8:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0020(2) OFF
34 9:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0021(2) OFF
35 10:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x0022(2) OFF
36 11:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x0023(2) OFF
37 12:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0028(3) OFF
38 13:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0029(3) OFF
39 14:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x002a(3) OFF
40 15:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x002b(3) OFF
41 16:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0040(4) OFF
42 17:0 --:UP 100G 62:69 0x00f0:0x0030 0:1 0:1 0x0041(4) OFF
43 18:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x0042(4) OFF
44 19:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x0043(4) OFF
45 20:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0048(5) OFF
46 21:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0049(5) OFF
47 22:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x004a(5) OFF
48 23:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x004b(5) OFF
49 24:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0060(6) OFF
50 25:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0061(6) OFF
51 26:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x0062(6) OFF
52 27:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x0063(6) OFF
53 28:0 --:UP 100G 63:69 0x000f:0x0003 0:1 0:1 0x0068(7) OFF
54 29:0 --:UP 100G 63:69 0x00f0:0x0030 0:1 0:1 0x0069(7) OFF
55 30:0 --:UP 100G 63:69 0x0f00:0x0300 0:1 0:1 0x006a(7) OFF
56 31:0 --:UP 100G 63:69 0xf000:0x3000 0:1 0:1 0x006b(7) OFF
57 32:0 --:UP 400G 79:79 0x00ff:0x00ff 0:1 0:1 0x0100(8) OFF
58 33:0 --:UP 400G 79:79 0xff00:0xff00 0:1 0:1 0x0102(8) OFF
59 34:0 --:UP 400G 79:79 0x00ff:0x00ff 0:1 0:1 0x0108(9) OFF
60 35:0 --:UP 400G 79:79 0xff00:0xff00 0:1 0:1 0x010a(9) OFF
61 -----
62 L:Line H:Host SD:Signal detect LOCK:Dsp lock !e: disable
63
64 regress@rtme-acx7100-32C:pfe> show system info
65
66 System Date : Mon Jul 11 07:56:30 PDT 2022
67 System Uptime : Up for 38-15:21:05
68 Pfe Uptime : Up for 38-15:19:39
69
70 CPU INFO:
71 =====
72 Model name : Intel(R) Xeon(R) CPU D-1637 @ 2.90GHz
73 Configured CPU Speed : 1956.403 2730.638 1738.774 2876.145 2279.032 1903.599 2392.586 2759.427
74 2710.492 2428.014 1909.381 2113.616 MHz
75 Physical cores : 6
76 Hyperthreaded cores : 12
77 L1 Cache size : 0 kB
78 L2 Cache size : 0 kB
79 L23Cache size : 0 kB
80
81 CPU PAIRS:
82 =====
83 Core 0 : 0 6
84 Core 1 : 1 7
85 Core 2 : 2 8
86 Core 3 : 3 9
87 Core 4 : 4 10
88 Core 5 : 5 11
89
90 SYSTEM MEMORY USAGE (MB):
91 =====
92 MemTotal: 65281228 kB
93 MemFree: 46410328 kB
94 MemAvailable: 59100436 kB
95 Buffers: 647040 kB
96 Cached: 12470188 kB
97 SwapCached: 0 kB
98 SwapTotal: 4194300 kB
99 SwapFree: 4194300 kB
100 <SNIP>

```

The ACX7100-48L differs by the absence of gearboxes, the ports being directly connected to the PFE Port Macros (hence the lack of MACsec support on this platform).

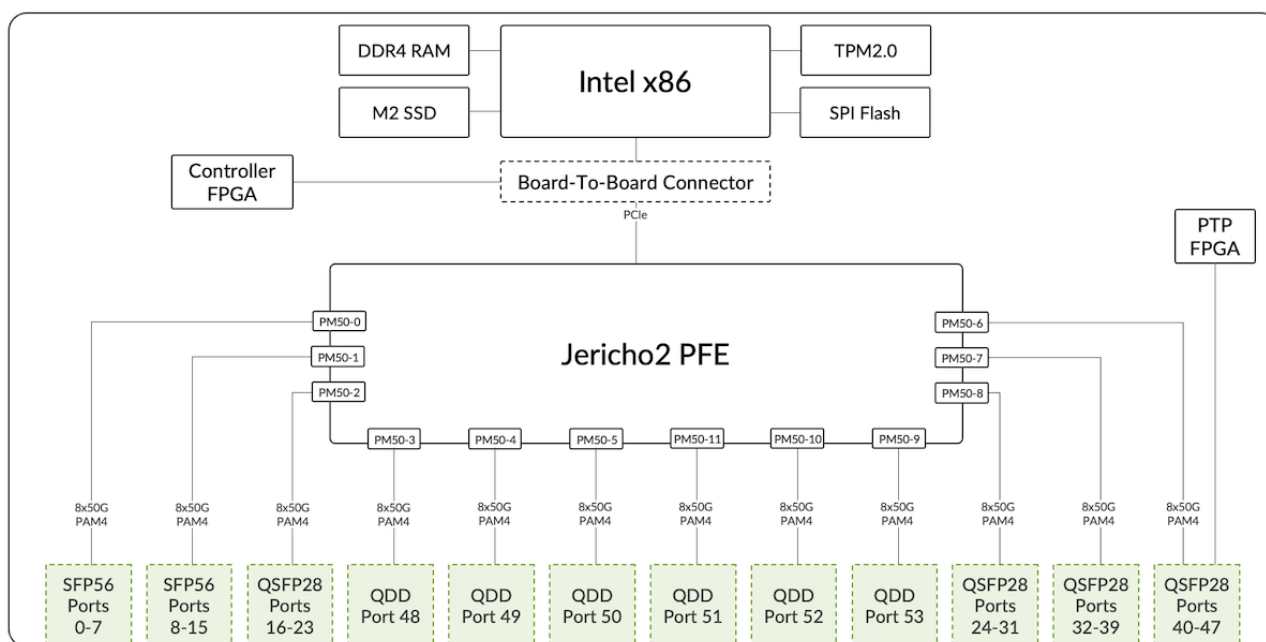


Figure 17: Internal Architecture

1.4 Fan Tray and Power Module Redundancy

Both ACX7100-48L and ACX7100-32C have a single routing engine. That means we don't support any kind of High-Availability (HA) option at the RE level. Also, In-Service System Upgrade (ISSU) is not supported, nor planned.

The only field replaceable parts on these routers are:

- the Power Supply Modules (2x AC or 2x DC)
- the Fan Trays (6 blocks)
- the SATA SDD

The SSD maintenance is well documented in the TechLibrary articles:

- <https://www.juniper.net/documentation/us/en/hardware/acx7100-48L/topics/topic-map/ssd-acx7100-48L-remove-install.html>
- <https://www.juniper.net/documentation/us/en/hardware/acx7100-32C/topics/topic-map/ssd-acx7100-32C-remove-install.html>

Regarding the cooling system and power supply, they are all accessible from the back of the router:

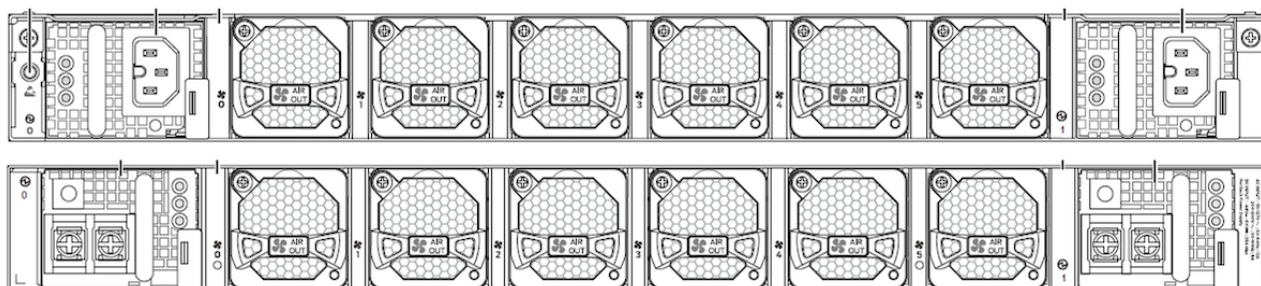


Figure 18: AC and DC Systems Rear View

You notice two PSMs, one on each side of the router. They can be AC or DC but we can't mix the two categories. With a 1+1 PSM configuration, we have power redundancy: if one fails, the system raises an alarm but stays fully operational.

```

1 regress@rtme-acx7100-32C> show chassis power detail
2
3 Chassis Power Voltage(V) Power(W)
4
5 Total Input Power 468
6   PSM 0
7     State: Online
8     Input 202 223
9     Output 11.97 201.25
10    Capacity 1600 W (maximum 1600 W)
11   PSM 1
12     State: Online
13     Input 202 245
14     Output 11.95 221.57
15    Capacity 1600 W (maximum 1600 W)
16
17 System:
18   Zone 0:
19     Capacity: 3200 W (maximum 3200 W)
20     Actual usage: 468 W
21     Total system capacity: 3200 W (maximum 3200 W)
22
23 regress@rtme-acx7100-32C>

```

ACX7100 relies on six Fan Trays for cooling purposes:

- ACX7100-48L is designed for both front-to-back airflow and back-to-front airflow
- ACX7100-32C is designed only for front-to-back airflow

Each Fan Tray is a pluggable module hosting two 40x40 mm counter-rotating rotors (Fan 1 and Fan 2 below). The system supports single rotor failure (N+1 redundancy).

```

1 regress@rtme-acx7100-32C> show chassis fan
2
3   Item Status % RPM Measurement
4   Fan Tray 0 Fan 1 OK 43% 12800 RPM
5   Fan Tray 0 Fan 2 OK 43% 10800 RPM
6   Fan Tray 1 Fan 1 OK 43% 13000 RPM
7   Fan Tray 1 Fan 2 OK 44% 11000 RPM
8   Fan Tray 2 Fan 1 OK 43% 12800 RPM
9   Fan Tray 2 Fan 2 OK 43% 10800 RPM
10  Fan Tray 3 Fan 1 OK 43% 12800 RPM
11  Fan Tray 3 Fan 2 OK 43% 10800 RPM
12  Fan Tray 4 Fan 1 OK 43% 13000 RPM
13  Fan Tray 4 Fan 2 OK 43% 10800 RPM
14  Fan Tray 5 Fan 1 OK 43% 12800 RPM
15  Fan Tray 5 Fan 2 OK 44% 11000 RPM
16
17 regress@rtme-acx7100-32C>

```

Note: it is possible to deactivate half of the ports and shut down the associated PFE core. This power-saving mode can reduce the consumption for low port density deployments: <https://www.juniper.net/documentation/us/en/software/junos/chassis/topics/task/configuring-power-saving-mode.html>

1.5 PFE Description

We invite you to check the articles to learn more about the ASIC selection process and many more details on the packet forwarding ASIC.

- Building the ACX7000 Series: the PFE: <https://juniper.github.io/techposts/building-the-acx7000-series-the-pfe/article>

- Let's Talk About VOQ and DNX Pipeline: <https://juniper.github.io/techposts/voq-and-dnx-pipeline/article>

The ACX7100-48L and ACX7100-32C are “Systems on the Chip” (SoC). That means they are based on a single PFE where all parts involved in the forwarding tasks are directly connected. This Jericho2 (BCM88690) PFE is a deep-buffer ASIC with hybrid packet memory (internal and external). The external buffer is actually present in the package itself: 8GB of HBM2 memory.

Follow-up articles in the near future will describe some internal key aspects of the PFE like the MDB profiles and the resource allocation/monitoring.

1.6 MACsec and Timing

MACsec is only supported on the ACX7100-32C ports (all of them) and not on the ACX7100-48L. The feature is fully handled by the gearbox.

Follow-up articles will be dedicated to security functions (including MACsec) and timing.

1.7 In Conclusion

The ACX7100-48L and ACX7100-32C routers are 1-RU, 4.8Tbps platforms, ideally positioned for multiple roles including 10G/25/50GE and 100GE to 400GE aggregation. In the next articles, we will continue the description of the ACX7000 portfolio and dig deeper into the specific features of these products.

1.8 Useful links

- Building the ACX7100 Series: the PFE: <https://juniper.github.io/techposts/building-the-acx7000-series-the-pfe/article>
- Let's talk about VOQ and DNX Pipeline: <https://juniper.github.io/techposts/voq-and-dnx-pipeline/article>
- Datasheet: <https://www.juniper.net/content/dam/www/assets/datasheets/us/en/routers/acx7100-cloud-metro-routers-datasheet.pdf>
- Hardware Compatibility Tool:

<https://apps.juniper.net/hct/product/?prd=ACX7100-48L> <https://apps.juniper.net/hct/product/?prd=ACX7100-32C>

- Hardware guides:

<https://www.juniper.net/documentation/us/en/hardware/acx7100-48L/topics/topic-map/acx7100-48L-system-overview.html> <https://www.juniper.net/documentation/us/en/hardware/acx7100-32C/topics/topic-map/acx7100-32C-system-overview.html> - SSD Maintenance

<https://www.juniper.net/documentation/us/en/hardware/acx7100-48L/topics/topic-map/ssd-acx7100-48L-remove-install.html> <https://www.juniper.net/documentation/us/en/hardware/acx7100-32C/topics/topic-map/ssd-acx7100-32C-remove-install.html>

1.9 Glossary

- AFI: AirFlow In
- AFO: AirFlow Out
- FRS: First Revenue Ship
- GB: GearBox
- HBM: High Bandwidth Memory
- I2C: Inter-Integrated Circuit
- NRZ: Non-Return to Zero
- PAM4: Pulse Amplification Modulation 4
- PFE: Packet Forwarding Engine
- PG: Port Group
- PM: Port Macro
- PSM : Power Supply Module
- PTP: Precision Time Protocol
- QSA : QSFP to SFP Adaptor
- SerDes: Serializer Deserializer
- SKU: Stock Keeping Unit
- SSD / SATA: Solid State Drive / Serial Advanced Technology Attachment
- TPM : Trusted Platform Module

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